

Central Line-Associated Bloodstream Infection (CLABSI) Prevention

Policy Number	POL-IC-021	Owner / Department	Infection Prevention & Control / Nursing
Effective Date	2023-08-15	Version	3.2
Last Revised	2026-02-18	Next Review	2028-02-18
Approved By	Infection Prevention & Control Committee	Applies To	All units and providers who insert, access, or maintain central venous catheters, including ICUs, procedural areas, and oncology

1. Purpose

To eliminate central line-associated bloodstream infections through standardized insertion and maintenance practices, consistent with the CDC/HICPAC Guidelines for the Prevention of Intravascular Catheter-Related Infections and the SHEA/IDSA/APIC Compendium.

2. Definitions

- **Central line** — an intravascular catheter that terminates at or close to the heart or in a great vessel and is used for infusion, withdrawal of blood, or hemodynamic monitoring.
- **CLABSI** — a laboratory-confirmed bloodstream infection in a patient with a central line in place for > 2 calendar days on the date of the event, not related to an infection at another site, per the NHSN surveillance definition.
- **Maximal sterile barrier** — cap, mask, sterile gown, sterile gloves, and a full-body sterile drape used during insertion.
- **Central line day** — the daily count of patients with one or more central lines, the denominator for the NHSN CLABSI rate.

3. Insertion and Maintenance Bundles

Both bundles are all-or-nothing: every element is required for every line. An independent observer completes the insertion checklist and is empowered to halt a non-emergent insertion if any element is breached.

Bundle	Required elements
Insertion	Hand hygiene; maximal sterile barrier precautions; chlorhexidine > 0.5% with alcohol skin prep allowed to fully dry; avoid the femoral site in adults; checklist completed by an observer.
Maintenance	Daily review of line necessity with prompt removal; scrub the hub for 15 seconds before every access; sterile dressing change (gauze every 2 days, transparent every 7 days or when soiled); CHG bathing for ICU patients.

IMPORTANT — Assess the need for every central line at least once daily and remove it as soon as it is no longer essential — an unnecessary catheter day is the single largest modifiable CLABSI risk. Any line inserted under emergent, non-sterile conditions must be replaced within 48 hours.

4. Severity & Reporting

Each confirmed CLABSI is an event under POL-RM-013. A CLABSI resulting in sepsis, septic shock, ICU transfer, or death is SEV-1; the attending provider, Charge Nurse, Nursing Supervisor, Risk Management, Infection Prevention, and the Administrator-on-Call must be notified immediately and never later than 1 hour from discovery. A CLABSI requiring antimicrobial therapy or line removal without those complications is SEV-2 — notify the attending provider, Charge Nurse, Risk Management, and Infection Prevention within 4 hours. An insertion-bundle breach caught before harm is a Near Miss, reported within 24 hours.

Severity	CLABSI event	Notify by
SEV-1	CLABSI with sepsis/septic shock, ICU transfer, or death.	≤ 1 hour from discovery.
SEV-2	Confirmed CLABSI requiring treatment or line removal, no life-threatening sequelae.	≤ 4 hours.
SEV-3 / Near Miss	Bundle breach or line issue intercepted before infection.	≤ 24 hours.

5. Documentation Requirements

- Completed insertion checklist with site, catheter type, number of lumens, and observer name.
- Daily line-necessity assessment and dressing/hub-care records.
- For a confirmed CLABSI: organism, date of positive culture, central line days, assigned severity, and all notifications; reported to NHSN by Infection Prevention.

6. References

- CDC/HICPAC, Guidelines for the Prevention of Intravascular Catheter-Related Infections (2011, with updates).
- SHEA/IDSA/APIC, Strategies to Prevent CLABSI in Acute Care Hospitals (2022 Update).
- CDC NHSN, Bloodstream Infection Event (CLABSI) surveillance definition, Patient Safety Component Manual.
- Related: POL-RM-013 (severity & reporting), POL-IC-020 (hand hygiene).